EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. The intercept made by the plane $\vec{r}.\vec{n}=q$ on the x-axis is

(A) $\frac{q}{\hat{i}.\vec{n}}$ (B) $\frac{\hat{i}.\vec{n}}{q}$ (C) $\left(\hat{i}.\vec{n}\right)q$ (D) $\frac{q}{|\vec{n}|}$

MTD142

2. If the distance between the planes
$8x + 12y - 14z = 2$
and $4x + 6y - 7z = 2$
can be expressed in the form $\frac{1}{\sqrt{N}}$ where N is natural then the value of $\frac{N(N+1)}{2}$ is

(A) 4950 (B) 5050 (C) 5150 (D) 5151

MTD143

3. A plane passes through the point $P(4, 0, 0)$ and $Q(0, 0, 4)$ and is parallel to the y-axis. The distance of the plane from the origin is

(A) 2 (B) 4 (C) $\sqrt{2}$ (D) $2\sqrt{2}$

MTD144

4. If from the point $P(f, g, h)$ perpendiculars PL, PM be drawn to yz and zx planes then the equation to the plane OLM is

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(A) $\frac{x}{f}+\frac{y}{g}-\frac{z}{h}=0$ (B) $\frac{x}{f}+\frac{y}{g}+\frac{z}{h}=0$ (C) $\frac{x}{f}-\frac{y}{g}+\frac{z}{h}=0$ (D) $-\frac{x}{f}+\frac{y}{g}+\frac{z}{h}=0$

MTD145

5. If the plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}(k)$ with x-axis, then k is equal to

(A) $\sqrt{3}/2$ (B) $2/7$ (C) $\sqrt{2}/3$ (D) 1

MTD146

6. The plane XOZ divides the join of $(1, -1, 5)$ and $(2, 3, 4)$ in the ratio $\lambda : 1$, then λ is

(A) – 3 (B) – 1/3 (C) 3 (D) 1/3

MTD147

7. A variable plane forms a tetrahedron of constant volume $64\ K^3$ with the coordinate planes and the origin, then locus of the centroid of the tetrahedron is

(A) $x^3 + y^3 + z^3 = \pm 6K^3$ (B) $xyz = \pm\ 6K^3$
(C) $x^2 + y^2 + z^2 = 4K^2$ (D) $x^{-2} + y^{-2} + z^{-2} = 4K^{-2}$

MTD148

8. Let $ABCD$ be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the area of triangles ABC, ACD and ADB be 3, 4 and 5 sq. units respectively. Then the area of the triangle BCD, is

(A) $5\sqrt{2}$ (B) 5 (C) $5/\sqrt{2}$ (D) 5/2

MTD149

For More Join: @IITJEE_Advanced

9. Equation of the line which passes through the point with p. v. (2, 1, 0) and perpendicular to the plane containing the vectors $\hat{i}+\hat{j}$ and $\hat{j}+\hat{k}$ is

(A) $\vec{r}$ = (2, 1, 0) + t (1, –1, 1) (B) $\vec{r}$ = (2, 1, 0) + t (–1, 1, 1)

(C) $\vec{r}$ = (2, 1, 0) + t (1, 1, –1) (D) $\vec{r}$ = (2, 1, 0) + t (1, 1, 1)

where t is a parameter

MTD150

10. Which of the following planes are parallel but not identical?

$P_1 : 4x - 2y + 6z = 3$

$P_2 : 4x - 2y - 2z = 6$

$P_3 : -6x + 3y - 9z = 5$

$P_4 : 2x - y - z = 3$

(A) P_2 & P_3 (B) P_2 & P_4 (C) P_1 & P_3 (D) P_1 & P_4

MTD151

11. A parallelopiped is formed by planes drawn through the points (1, 2, 3) and (9, 8, 5) parallel to the coordinate planes then which of the following is not the length of an edge of this rectangular parallelopiped

(A) 2 (B) 4 (C) 6 (D) 8

MTD152

12. Vector equation of the plane $\vec{r}=\hat{i}-\hat{j}+\lambda(\hat{i}+\hat{j}+\hat{k})+\mu(\hat{i}-2\hat{j}+3\hat{k})$ in the scalar dot product form is

(A) $\vec{r}.(5\hat{i}-2\hat{j}+3\hat{k})=7$ (B) $\vec{r}.(5\hat{i}+2\hat{j}-3\hat{k})=7$

(C) $\vec{r}.(5\hat{i}-2\hat{j}-3\hat{k})=7$ (D) $\vec{r}.(5\hat{i}+2\hat{j}+3\hat{k})=7$

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MTD153

13. The vector equations of the two lines L_1 and L_2 are given by

$L_1 : \vec{r}=2\hat{i}+9\hat{j}+13\hat{k}+\lambda(\hat{i}+2\hat{j}+3\hat{k})$; $L_2 : \vec{r}=-3\hat{i}+7\hat{j}+p\hat{k}+\mu(-\hat{i}+2\hat{j}-3\hat{k})$

then the lines L_1 and L_2 are

(A) skew lines for all $p\in R$

(B) intersecting for all $p\in R$ and the point of intersection is (–1, 3, 4)

(C) intersecting lines for p = – 2

(D) intersecting for all real $p\in R$

MTD154

14. Consider the plane (x, y, z) = (0, 1, 1) + λ(1, – 1, 1) + μ(2, – 1, 0). The distance of this plane from the origin is

(A) 1/3 (B) $\sqrt{3}/2$ (C) $\sqrt{3/2}$ (D) $2/\sqrt{3}$

MTD155

15. The value of 'a' for which the lines $\frac{x-2}{1}=\frac{y-9}{2}=\frac{z-13}{3}$ and $\frac{x-a}{-1}=\frac{y-7}{2}=\frac{z+2}{-3}$ intersect, is

(A) –5 (B) –2 (C) 5 (D) –3

MTD156

16. Given $A(1, -1, 0)$; $B(3, 1, 2)$; $C(2, -2, 4)$ and $D(-1, 1, -1)$ which of the following points neither lie on AB nor on CD?

(A) (2, 2, 4) (B) (2, –2, 4) (C) (2, 0,1) (D) (0, –2, –1)

MTD157

For More Join: @IITJEE_Advanced

17. For the line $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$, which one of the following is incorrect?

(A) it lies in the plane $x - 2y + z = 0$ (B) it is same as line $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$

(C) it passes through (2, 3, 5) (D) it is parallel to the plane $x - 2y + z - 6 = 0$

MTD158

18. Given planes $P_1 : cy + bz = x$

$P_2 : az + cx = y$

$P_3 : bx + ay = z$

P_1, P_2 and P_3 pass through one line, if

(A) $a^2 + b^2 + c^2 = ab + bc + ca$ (B) $a^2 + b^2 + c^2 + 2abc = 1$

(C) $a^2 + b^2 + c^2 = 1$ (D) $a^2 + b^2 + c^2 + 2ab + 2bc + 2ca + 2abc = 1$

MTD159

19. The line which contains all points (x, y, z) which are of the form $(x, y, z) = (2, -2, 5) + \lambda(1, -3, 2)$ intersects the plane $2x - 3y + 4z = 163$ at P and intersects the YZ plane at Q. If the distance PQ is $a\sqrt{b}$ where $a, b \in N$ and $a > 3$ then $(a + b)$ equals

(A) 23 (B) 95 (C) 27 (D) none

MTD160

20. Let L_1 be the line $\vec{r}_1 = 2\hat{i} + \hat{j} - \hat{k} + \lambda(\hat{i} + 2\hat{k})$ and let L_2 be the line $\vec{r}_2 = 3\hat{i} + \hat{j} + \mu(\hat{i} + \hat{j} - \hat{k})$.

Let Π be the plane which contains the line L_1 and is parallel to L_2. The distance of the plane Π from the origin is

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(A) $1/7$ (B) $\sqrt{2/7}$ (C) $\sqrt{6}$ (D) none

MTD161

21. The value of m for which straight line $3x - 2y + z + 3 = 0 = 4x - 3y + 4z + 1$ is parallel to the plane $2x - y + mz - 2 = 0$ is

(A) -2 (B) 8 (C) -18 (D) 11

MTD162

22. The distance of the point $(-1, -5, -10)$ from the point of intersection of the line $\frac{x-2}{2}=\frac{y+1}{4}=\frac{z-2}{12}$ and the plane $x - y + z = 5$ is

(A) $2\sqrt{11}$ (B) $\sqrt{126}$ (C) 13 (D) 14

MTD163

23. $P(\vec{p})$ and $Q(\vec{q})$ are the position vectors of two fixed points and $R(\vec{r})$ is the position vector of a variable point. If R moves such that $(\vec{r}-\vec{p})\times(\vec{r}-\vec{q})=\vec{0}$ then the locus of R is

(A) a plane containing the origin 'O' and parallel to two non collinear vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$

(B) the surface of a sphere described on PQ as its diameter.

(C) a line passing through the points P and Q

(D) a set of lines parallel to the line PQ.

MTD164

For More Join: @IITJEE_Advanced

24. Let the distance between the plane $3x + 4y + 5z + c_1 = 0$ and $3x + 4y + 5z - c_2 = 0$ is $2\sqrt{2}$, where $c_1, c_2 \in W$, then number of possible ordered pair(s) of (c_1, c_2) is -

(A) 1 (B) 21 (C) 31 (D) 51

MTD165

25. Consider the lines $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ and $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$, then the equation of the line which

(A) bisects the angle between the lines is $\frac{x}{3}=\frac{y}{3}=\frac{z}{8}$

(B) bisects the angle between the lines is $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$

(C) passes through origin and is perpendicular to the given lines is $x = y = -z$

(D) none of these

MTD166

26. A plane meets the coordinate axes in A, B, C and (α, β, γ) is the centroid of the triangle ABC, then the equation of the plane is

(A) $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=3$ (B) $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=1$ (C) $\frac{3x}{\alpha}+\frac{3y}{\beta}+\frac{3z}{\gamma}=1$ (D) $\alpha x + \beta y + \gamma z = 1$

MTD167

27. Equation of plane which passes through the point of intersection of lines $\frac{x-1}{3}=\frac{y-2}{1}=\frac{z-3}{2}$ and $\frac{x-3}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ and at greatest distance from the point (0, 0, 0) is:

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(A) $4x + 3y + 5z = 25$ (B) $4x + 3y + 5z = 50$

(C) $3x + 4y + 5z = 49$ (D) $x + 7y - 5z = 2$

MTD168

28. The non zero value of 'a' for which the lines $2x - y + 3z + 4 = 0 = ax + y - z + 2$ and $x - 3y + z = 0 = x + 2y + z + 1$ are co-planar is :

(A) –2 (B) 4 (C) 6 (D) 0

MTD169

29. A line having direction ratios 3, 4, 5 cuts 2 planes $2x - 3y + 6z - 12 = 0$ and $2x - 3y + 6z + 2 = 0$ at point P & Q, then find length of PQ

(A) $\frac{35\sqrt{2}}{12}$ (B) $\frac{35\sqrt{2}}{24}$ (C) $\frac{35\sqrt{2}}{6}$ (D) $\frac{35\sqrt{2}}{8}$

MTD170

30. A line L_1 having direction ratios 1, 0, 1 lies on xz plane. Now this xz plane is rotated about z-axis by an angle of 90°. Now the new position of L_1 is L_2. The angle between L_1 & L_2 is :

(A) 30° (B) 60° (C) 90° (D) 45°

MTD171

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Equation of plane containing the line $2x - y + z - 3 = 0 = 3x + y + z - 5$ and lying at a distance $\frac{1}{\sqrt{6}}$ units from the point (2,1,–1) can be-

(A) $2x - y + z - 3 = 0$ (B) $62x + 29y + 19z - 105 = 0$

(C) $2x + y - z - 5 = 0$ (D) $62x + 29y - 19z + 104 = 0$

MTD172

2. The volume of a right triangular prism $ABCA_1B_1C_1$ is equal to 3. If the position vectors of the vertices of the base ABC are A(1, 0, 1) ; B(2,0, 0) and C(0, 1, 0) then the position vectors of the vertex A_1 can be:

(A) (2, 2, 2) (B) (0, 2, 0) (C) (0, –2, 2) (D) (0, - 2, 0)

MTD173

3. Consider Lines $L_1 : \frac{x-\alpha}{1} = \frac{y}{-2} = \frac{z+\beta}{2}$, $L_2 : x = \alpha, \frac{y}{-\alpha} = \frac{z+\alpha}{2-\beta}$, plane $P : 2x + 2y + z + 7 = 0$. Let line L_2 lies in plane P, then

(A) $\alpha = -7$

(B) $\alpha = 7$

(C) minimum distance between line L_1 and plane P is 11.

(D) minimum distance between line L_1 and plane P is $\frac{23}{3}$

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MTD174

4. The point $\vec{A}$ (3,4,7), $\vec{B}$ (4,5,9) and $\vec{C}$ (1,2,–1) are three vertices of a parallelogram $ABCD$, then -

(A) vector equation of line AB is $\vec{r} = 4\hat{i} + 5\hat{j} + 9\hat{k} + \lambda\left(\hat{i} + \hat{j} + 2\hat{k}\right)$

(B) cartesian equation of line BC is $\frac{x-4}{3} = \frac{y-5}{3} = \frac{z-9}{10}$

(C) coordinates of D are (0,1,–3)

(D) $ABCD$ is rectangle

MTD175

5. Let two planes $P_1 : 2x - y + z - 2 = 0$ and $P_2 : x + 2y - z - 3 = 0$ are given then -

(A) The equation of the plane through line of intersection of $P_1 = 0$ and $P_2 = 0$ and the point (3,2,1) is $x - 3y + 2z + 1 = 0$

(B) The equation of the plane through line of intersection of $P_1 = 0$ and $P_2 = 0$ and the point (3,2,1) is $3x - y + 2z - 9 = 0$

(C) The equation of acute angle bisector plane of $P_1 = 0$ and $P_2 = 0$ is $x - 3y + 2z + 1 = 0$

(D) The equation of acute angle bisector plane of $P_1 = 0$ and $P_2 = 0$ is $x + 3y + 2z + 2 = 0$

MTD176

6. A variable point P(4 cost, 4sint, 4sint) moves in space, now which of the following holds good ?

(A) Point P moves on a plane (B) Point 'P' traces a circle.

(C) Area enclosed by P is $16\sqrt{2}\pi$ (D) Point P cannot lie on a fixed plane.

MTD177

For More Join: @IITJEE_Advanced

7. The projection of line $\frac{x}{2}=\frac{y-1}{2}=\frac{z-1}{1}$ on a plane 'P' is $\frac{x}{1}=\frac{y-1}{1}=\frac{z-1}{-1}$. If the plane P passes through $(k, -2, 0)$, then k is greater than -

(A) 2 (B) 3 (C) 5 (D) 4

MTD178

8. A line segment has length 6 and direction ratios are –3,4,6, then the component of the line vector are-

(A) $\frac{-18}{\sqrt{61}},\frac{24}{\sqrt{61}},\frac{36}{\sqrt{61}}$ (B) 27,–18,–54

(C) 27, –18,54 (D) $\frac{18}{\sqrt{61}},\frac{-24}{\sqrt{61}},\frac{-36}{\sqrt{61}}$

MTD179

9. Which of the following is (are) correct -

(A) If two lines in space are not intersecting, then they must be skew lines.

(B) If two lines are parallel to a plane 'P', then their direction ratios will be proportional

(C) If two lines are perpendicular to a plane 'P', then their direction ratios will be proportional

(D) Equation $\frac{x+1}{a}=\frac{y-1}{b}=\frac{z}{c}$, where a, b, c are real parameters, represents a family of concurrent lines in space

MTD180

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10. Given the equations of the line $3x - y + z + 1 = 0, 5x + y + 3z = 0$.
Then which of the following is correct ?

(A) Symmetrical form of the equations of line is $\frac{x}{2}=\frac{y-\frac{1}{8}}{-1}=\frac{z+\frac{5}{8}}{1}$

(B) symmetrical form of the equations of line is $\frac{x+\frac{1}{8}}{1}=\frac{y-\frac{5}{8}}{1}=\frac{z}{-2}$

(C) equation of the plane through (2, 1, 4) and perpendicular to the given lines is $2x - y + z - 7 = 0$

(D) equation of the plane through (2, 1, 4) and perpendicular to the given lines is $x + y - 2z + 5 = 0$

MTD181

11. Consider the family of planes $x + y + z = c$ where is a parameter intersecting the coordinate axes at P, Q, R and α, β, γ are the angles made by each member of this family with positive x, y and z axis. Which of the following interpretations hold good for this family -

(A) each member of this family is equally inclined with the coordinate axes.

(B) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$

(C) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2$

(D) for $c = 3$ area of the triangle PQR is $3\sqrt{3}$ sq. units.

MTD182

For More Join: @IITJEE_Advanced

12. Consider a plane P passing through $A(\lambda,3,\mu)$, $B(-1,3,2)$ and $C(7,5,10)$ and a straight line L with positive direction cosines passing through A, bisecting BC and makes equal angles with the coordinate axes. Let L_1 be a line parallel to L and passing through origin. Which of the following is(are) correct?

(A) The value of $(\lambda + \mu)$ is equal to 5.

(B) Equation of straight line L_1 is $\frac{x-1}{1}=\frac{y-1}{1}=\frac{z-1}{1}$.

(C) Equation of the plane perpendicular to the plane P and containing line L_1 is $x - 2y + z = 0$

(D) Area of triangle ABC is equal to $3\sqrt{2}$.

MTD183

13. A line L passing through the point $P(1,4,3)$, is perpendicular to both the lines $\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4}$ and $\frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$.

If the position vector of point Q on L is(a_1,a_2,a_3) such that $(PQ)^2 = 357$, then $(a_1 + a_2 + a_3)$ can be-

(A) 16 (B) 15 (C) 2 (D) 1

MTD184

14. Which of the following is (are) correct -

(A) If two lines in space are not intersecting, then they must be skew lines.

(B) If two lines are parallel to a plane 'P', then their direction ratios will be proportional

(C) If two lines are perpendicular to a plane 'P', then their direction ratios will be proportional

(D) Equation $\frac{x+1}{a}=\frac{y-1}{b}=\frac{z}{c}$, where a, b, c are real parameters, represents a family of concurrent lines in space

MTD185

15. The direction cosines of the lines bisecting the angle between the lines whose direction cosines are ℓ_1, m_1, n_1 and ℓ_2, m_2, n_2 and the angle between these lines is θ, are

(A) $\frac{\ell_1-\ell_2}{\cos\frac{\theta}{2}},\frac{m_1-m_2}{\cos\frac{\theta}{2}},\frac{n_1-n_2}{\cos\frac{\theta}{2}}$ (B) $\frac{\ell_1+\ell_2}{2\cos\frac{\theta}{2}},\frac{m_1+m_2}{2\cos\frac{\theta}{2}},\frac{n_1+n_2}{2\cos\frac{\theta}{2}}$

(C) $\frac{\ell_1+\ell_2}{2\sin\frac{\theta}{2}},\frac{m_1+m_2}{2\sin\frac{\theta}{2}},\frac{n_1+n_2}{2\sin\frac{\theta}{2}}$ (D) $\frac{\ell_1-\ell_2}{2\sin\frac{\theta}{2}},\frac{m_1-m_2}{2\sin\frac{\theta}{2}},\frac{n_1-n_2}{2\sin\frac{\theta}{2}}$

MTD186

16. The vertices of a triangle are A (1, 1, 2), B(4, 3, 1) and C(2, 3, 5). A vector representing the bisector of the angle A is :

(A) $2\hat{i}-4\hat{k}$ (B) $-2\hat{i}+4\hat{k}$ (C) $-2\hat{i}-2\hat{j}-\hat{k}$ (D) $2\hat{i}+2\hat{j}+\hat{k}$

MTD187

For More Join: @IITJEE_Advanced

17. Acute angle between the lines $\frac{x-1}{\ell}=\frac{y+1}{m}=\frac{z}{n}$ and $\frac{x+1}{m}=\frac{y-3}{n}=\frac{z-1}{\ell}$ where $\ell > m > n$, and ℓ, m, n are the roots of the cubic equation $x^3 + x^2 - 4x = 4$ is equal to :

(A) $\cos^{-1}\frac{3}{\sqrt{13}}$ (B) $\sin^{-1}\frac{\sqrt{65}}{9}$ (C) $2\cos^{-1}\sqrt{\frac{13}{18}}$ (D) $\tan^{-1}\frac{2}{3}$

MTD188

18. The line $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-1}{-1}$ intersects the curve $xy = c^2, z = 0$ if c is equal to :

(A) –1 (B) $-\sqrt{5}$ (C) $\sqrt{5}$ (D) 1

MTD189

19. Three distinct lines $\frac{x-1}{3}=\frac{y-2}{2}=\frac{z-3}{1}$, $\frac{x-1}{5}=\frac{2y-4}{3}=\frac{3z-9}{1}$, $\frac{x-\lambda^2}{3}=\frac{y-2}{2}=\frac{z-3}{\lambda}$ are concurrent the value of λ may be :

(A) 1 (B) –1 (C) 2 (D) –2

MTD190

20. The line $\frac{x+6}{5}=\frac{y+10}{3}=\frac{z+14}{8}$ is the hypotenuse of an isosceles right angle triangle whose opposite vertex is (7, 2, 4) Then the equation of remaining sides is/are -

(A) $\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}$ (B) $\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6}$

(C) $\frac{x+7}{3}=\frac{y+2}{6}=\frac{z+4}{2}$ (D) $\frac{x+7}{2}=\frac{y+2}{-3}=\frac{z+4}{6}$

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MTD191

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 21 to 22

Consider planes $P_1 : x + 2y + z - 1 = 0$ & $P_2 : x - y + 2z + 1 = 0$ and a plane P containing the line of intersection of P_1 & P_2.

21. If P is perpendicular to the plane $x + y + z = 5$, then equation of P is -

(A) $2x - 4y + 3z = 3$ (B) $x + 4y + z = 3$

(C) $x - 4y + 3z + 3 = 0$ (D) $-x - 4y + 4z + 3 = 0$

MTD192

22. If from a point Q lying on plane P in Ist octant the distances of planes $x = 0, y = 0$ & $z = 0$ are 1, 1 and 2 respectively, then vector equation of line passing through origin and parallel to normal of the plane P is -

(A) $\vec{r}=\lambda(\hat{i}-14\hat{j}+2\hat{k})$ (B) $\vec{r}=\lambda(2\hat{i}-\hat{j}+\hat{k})$

(C) $\vec{r}=\lambda(\hat{i}-\hat{j}+\hat{k})$ (D) $\vec{r}=\lambda(\hat{i}+14\hat{j}-3\hat{k})$

(where $\lambda \in R$)

MTD193

For More Join: @IITJEE_Advanced

Paragraph for Question No. 23 to 24

Consider a plane $\Pi : \vec{r}.\left(\hat{i}-\hat{j}+2\hat{k}\right)=5$, a line $L_1 : \vec{r}=\left(2\hat{i}-\hat{j}+\hat{k}\right)+\lambda\left(4\hat{i}+\hat{j}-\hat{k}\right)$ and a point $A(2,2,1)$.

L_2 is a line passing through A intersecting L_1 and parallel to plane Π -

23. Equation of L_2 is -

(A) $\vec{r}=(2+6\lambda)\hat{i}-2\hat{j}+\lambda\hat{k}$

(B) $\vec{r}=2\hat{i}-(2+6\lambda)\hat{j}+\hat{k}$

(C) $\vec{r}=(4\lambda+6)\hat{i}+2\lambda\hat{j}-\lambda\hat{k}$

(D) $\vec{r}=(2+6\lambda)\hat{i}+(\lambda-3)\hat{j}+(2+\lambda)\hat{k}$

MTD194

24. Line L_1 intersects plane Π at Q and xy-plane at R, then volume of tetrahedron $OAQR$ is (where O is origin)

(A) 1 cubic units

(B) $\frac{1}{2}$ cubic units

(C) 6 cubic units

(D) 4 cubic units

MTD195

Paragraph for Question No. 25 to 26

A conic section is obtained by the intersection of two inverted cones (having same vertex) with a plane. If a plane passes through the vertex, then its intersection with the cones either represents a pair of straight lines or a point depending upon whether it intersects the cones or not. If plane does not pass through vertex, then section may be hyperbola or circle depending upon whether plane is parallel to axis of the cone or perpendicular to it. If it has any other inclination, then intersection of plane and cone gives an ellipse or a parabola.

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25. If a variable point P moves such that the line passing through P and $Q(0,0,2)$ makes an angle θ with z-axis, then the locus of P is :

(A) z-axis if $\theta = 0°$

(B) Plane $z = 2$ if $\theta = 90°$

(C) A cone $x^2 + y^2 - 3(z-2)^2 = 0$ if $\theta = 60°$

(D) A cone $x^2 + y^2 + 3(z-2)^2 = 0$ if $\theta = 60°$

MTD196

26. In the above question if $\theta = 60°$ then the locus of intersection of locus of P and $ax + y + z = b$ will be

(A) a hyperbola if $a = b = 1$

(B) a pair of straight line if $a = 1$ and $b = 2$

(C) a pair of straight lines if $a = b = 2$

(D) none of these

MTD197

Paragraph for Question No. 27 to 28

In 3D geometry, a tetrahedron is a four-faced polyhedron with four vertices. It is one of the simplest and most important shapes in 3D geometry, and it has many practical applications in fields such as engineering, physics, chemistry and computer graphics. To solve problems related to a tetrahedron, we need to use concepts such as shortest distance formula, cross product, and the formula for the volume of a tetrahedron. By applying these concepts, we can gain a better understanding of 3D geometry and solve problems related to the tetrahedron.

27. Find the shortest distance between any two opposite edges of a tetrahedron formed by the planes $y + z = 0, x + z = 0, x + y = 0, x + y + z = \sqrt{3}$.

MTD198

For More Join: @IITJEE_Advanced

28. If the volume of tetrahedron formed by planes whose equations are $y + z = 0$, $z + x = 0$, $x + y = 0$ and $x + y + z = 1$ is t cubic unit, then the value of t is

MTD199

MATCHING LIST TYPE QUESTION

29. Consider the following four pairs of lines in **List-I** and match them with one or more entries in **List-II**.

	List-I	**List-II**
(I)	$L_1 : x = 1 + t,\ y = t,\ z = 2 - 5t$ $L_2 : \vec{r} = (2,1,-3) + \lambda(2, 2, -10)$	(P) non coplanar lines

MTD200

(II)	$L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$ $L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$	(Q) lines lie in a unique plane

MTD201

(III)	$L_1 : x = -6t, y = 1 + 9t, z = -3t$ $L_2 : x = 1 + 2s,\ y = 4 - 3s,\ z = s$	(R) infinite planes containing both the lines

MTD202

(IV)	$L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ $L_2 \frac{x-3}{-4} : = \frac{y-2}{-3} = \frac{z-1}{2}$	(S) lines are not intersecting at a unique point

MTD203

(A) I $\rightarrow$ Q,S; II $\rightarrow$ P,S; III $\rightarrow$ R,S; IV $\rightarrow$ Q
(B) I $\rightarrow$ Q; II $\rightarrow$ Q,S; III $\rightarrow$ P,S; IV $\rightarrow$ R,S
(C) I $\rightarrow$ P,S; II $\rightarrow$ Q,S; III $\rightarrow$ Q; IV $\rightarrow$ R,S
(D) I $\rightarrow$ R,S; II $\rightarrow$ Q; III $\rightarrow$ Q,S; IV $\rightarrow$ P,S

MATRIX MATCH TYPE QUESTION

30. $P(0, 3, -2)$; $Q(3, 7, -1)$ and $R(1, -3, -1)$ are 3 given points. Let L_1 be the line passing through P and Q and L_2 be the line through R and parallel to the vector $\vec{V} = \hat{i} + \hat{k}$.

	Column-I		**Column-II**
(A)	perpendicular distance of P from L_2	(P)	$7\sqrt{3}$
(B)	shortest distance between L_1 and L_2	(Q)	2
(C)	area of the triangle PQR	(R)	6
(D)	distance from (0, 0, 0) to the plane PQR	(S)	$\frac{19}{\sqrt{147}}$

MTD204

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EXERCISE (S)

1. For what value of p the lines $\frac{x+1}{-3}=\frac{y-p}{2}=\frac{z+2}{1}$ and $\frac{x}{1}=\frac{y-7}{-3}=\frac{z+7}{2}$ are in the same plane.

MTD114

2. The equation of the plane which has the property that the point $Q(5,4,5)$ is the reflection of point $P(1,2,3)$ through that plane, is $ax+by+cz=d$ where $a,b,c,d \in N$. Find the least value of $(a+b+c+d)$.

MTD115

3. A line from origin meets the lines $\frac{x-1}{1}=\frac{y}{2}=\frac{z+1}{2}$ and $\frac{x}{2}=\frac{y-1}{3}=\frac{z}{1}$ at A and B, then length of AB is equal to

MTD116

4. Let the equation of locus of a point which moves such that its distance from $(1,1,1)$ is double its distance from x axis is $x^2-\lambda_1 y^2-\lambda_1 z^2-2x-2y-2z+\lambda_3=0$, then $|\lambda_1|+|\lambda_2|+|\lambda_3|$ is equal to

MTD117

5. If the distance between the plane $Ax-2y+z=d$ and the plane containing the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ is $\sqrt{6}$, then $|d|$ is

MTD118

6. If the distance between the plane, $23x-10y-2z+48=0$ and the plane containing the lines $\frac{x+1}{2}=\frac{y-3}{4}=\frac{z+1}{3}$ and $\frac{x+3}{2}=\frac{y+2}{6}=\frac{z-1}{\lambda}(\lambda \in R)$ is equal to $\frac{k}{\sqrt{633}}$, then k is equal to _____.

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MTD119

7. If the lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-1}{4}$ and $\frac{x-3}{1}=\frac{y-k}{2}=\frac{z}{1}$ intersect, then k is equal to:

MTD120

8. Distance between two parallel planes $2x+y+2z=8$ and $4x+2y+4z+5=0$ is :-

MTD121

9. If the foot of the perpendicular drawn from the point (1, 0, 3) on a line passing through (α, 7, 1) is $\left(\frac{5}{3},\frac{7}{3},\frac{17}{3}\right)$, then α is equal to____

MTD122

10. A line from the origin meets the lines $\frac{x-2}{1}=\frac{y-1}{-2}=\frac{z+1}{1}$ and $\frac{x-\frac{8}{3}}{2}=\frac{y+3}{-1}=\frac{z-1}{1}$ at P and Q respectively. If length $PQ=d$, then d is

MTD123

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. If the lines $x = ay + b, z = cy + d$ and $x = a'z + b', y = c'z + d'$ are perpendicular, then:

[JEE (Main) 2019]

(1) $cc' + a + a' = 0$ (2) $aa' + c + c' = 0$ (3) $ab' + bc' + 1 = 0$ (4) $bb' + cc' + 1 = 0$

MTD026

2. A plane which bisects the angle between the two given planes $2x - y + 2z - 4 = 0$ and $x + 2y + 2z - 2 = 0$, passes through the point: **[JEE (Main) 2019]**

(1) (2,4,1) (2) (2, – 4, 1) (3) (1, 4, – 1) (4) (1, – 4, 1)

MTD027

3. If the foot of the perpendicular drawn from the point (1, 0, 3) on a line passing through (α, 7, 1) is $\left(\frac{5}{3}, \frac{7}{3}, \frac{17}{3}\right)$, then α is equal to____ **[JEE (Main) 2020]**

MTD028

4. The shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is

[JEE (Main) 2020]

(1) $\frac{7}{2}\sqrt{30}$ (2) $3\sqrt{30}$ (3) 3 (4) $2\sqrt{30}$

MTD029

5. If for some α and β in R, the intersection of the following three Planes

$x + 4y - 2z = 1$

$x + 7y - 5z = \beta$

$x + 5y + \alpha z = 5$

is a line in R^3, then α + β is equal to : **[JEE (Main) 2020]**

(1) 10 (2) –10 (3) 2 (4) 0

MTD030

6. Let P be a plane $lx + my + nz = 0$ containing the line, $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$. If plane P divides the line segment AB joining points $A(-3, -6, 1)$ and $B(2, 4, -3)$ in ratio $k : 1$ then the value of k is equal to : **[JEE (Main) 2021]**

(1) 1.5 (2) 3 (3) 2 (4) 4

MTD031

7. Let the plane $ax + by + cz + d = 0$ bisect the line joining the points $(4, -3, 1)$ and $(2, 3, -5)$ at the right angles. If a, b, c, d are integers, then the minimum value of $(a^2 + b^2 + c^2 + d^2)$ is

[JEE (Main) 2021]

MTD032

8. Consider the three planes

$P_1: 3x + 15y + 21z = 9$,

$P_2: x - 3y - z = 5$, and

$P_3: 2x + 10y + 14z = 5$

Then, which one of the following is true ? **[JEE (Main) 2021]**

(1) P_1 and P_2 are parallel (2) P_1 and P_3 are parallel

(3) P_2 and P_3 are parallel (4) P_1, P_2 and P_3 all are parallel

MTD033

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For More Join: @IITJEE_Advanced

9. The distance of the point (1, –2, 3) from the plane $x - y + z = 5$ measured parallel to a line, whose direction ratios are 2, 3, –6 is : **[JEE (Main) 2021]**

(1) 3 (2) 5 (3) 2 (4) 1

MTD034

10. Equation of a plane at a distance $\sqrt{\frac{2}{21}}$ from the origin, which contains the line of intersection of the planes $x - y - z - 1 = 0$ and $2x + y - 3z + 4 = 0$, is : **[JEE (Main) 2021]**

(1) $3x - y - 5z + 2 = 0$ (2) $3x - 4z + 3 = 0$

(3) $-x + 2y + 2z - 3 = 0$ (4) $4x - y - 5z + 2 = 0$

MTD035

11. Let a line having direction ratios $1, -4, 2$ intersect the lines $\frac{x-7}{3} = \frac{y-1}{-1} = \frac{z+2}{1}$ and $\frac{x}{2} = \frac{y-7}{3} = \frac{z}{1}$ at the point A and B. Then $(AB)^2$ is equal to ___ . **[JEE (Main) 2022]**

MTD036

12. Let the plane $2x + 3y + z + 20 = 0$ be rotated through a right angle about its line of intersection with the plane $x - 3y + 5z = 8$. If the mirror image of the point $\left(2, -\frac{1}{2}, 2\right)$ in the rotated plane is $B(a, b, c)$, then : **[JEE (Main) 2022]**

(1) $\frac{a}{8} = \frac{b}{5} = \frac{c}{-4}$ (2) $\frac{a}{4} = \frac{b}{5} = \frac{c}{-2}$ (3) $\frac{a}{8} = \frac{b}{-5} = \frac{c}{4}$ (4) $\frac{a}{4} = \frac{b}{5} = \frac{c}{2}$

MTD037

13. If two straight lines whose direction cosines are given by the relations $l + m - n = 0, 3l^2 + m^2 + cnl = 0$ are parallel, then the positive value of c is : **[JEE (Main) 2022]**

(1) 6 (2) 4 (3) 3 (4) 2

MTD038

14. The acute angle between the planes P_1 and P_2, when P_1 and P_2 are the planes passing through the intersection of the planes $5x + 8y + 13z - 29 = 0$ and $8x - 7y + z - 20 = 0$ and the points (2, 1, 3) and (0, 1, 2), respectively, is **[JEE (Main) 2022]**

(1) $\frac{\pi}{3}$ (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{12}$

MTD039

15. If the line of intersection of the planes $ax + by = 3$ and $ax + by + cz = 0, a > 0$ makes an angle 30° with the plane $y - z + 2 = 0$, then the direction cosines of the line are : **[JEE (Main) 2022]**

(1) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$ (2) $\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0$ (3) $\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, 0$ (4) $\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0$

MTD040

16. Let the lines $\frac{x-1}{\lambda} = \frac{y-2}{1} = \frac{z-3}{2}$ and $\frac{x+26}{-2} = \frac{y+18}{3} = \frac{z+28}{\lambda}$ be coplanar and P be the plane containing these two lines. Then which of the following points does **NOT** lies on P? **[JEE (Main) 2022]**

(1) (0, – 2, –2) (2) (– 5, 0, –1) (3) (3, –1, 0) (4) (0, 4, 5)

MTD041

For More Join: @IITJEE_Advanced

17. The foot of the perpendicular from a point on the circle $x^2+y^2=1, z=0$ to the plane $2x+3y+z=6$ lies on which one of the following curves? **[JEE (Main) 2022]**

(1) $(6x+5y-12)^2+4(3x+7y-8)^2=1, z=6-2x-3y$

(2) $(5x+6y-12)^2+4(3x+5y-9)^2=1, z=6-2x-3y$

(3) $(6x+5y-14)^2+9(3x+5y-7)^2=1, z=6-2x-3y$

(4) $(5x+6y-14)^2+9(3x+7y-8)^2=1, z=6-2x-3y$

MTD042

18. If (2, 3, 9), (5, 2, 1), (1, λ, 8) and (λ, 2, 3) are coplanar, then the product of all possible values of λ is: **[JEE (Main) 2022]**

(1) $\frac{21}{2}$ (2) $\frac{59}{8}$ (3) $\frac{57}{8}$ (4) $\frac{95}{8}$

MTD043

19. Let $P_1:\vec{r}.(2\hat{i}+j-3k)=4$ be a plane. Let P_2 be another plane which passes through the points $(2,-3,2)$ $(2,-2,-3)$ and $(1,-4,2)$. If the direction ratios of the line of intersection of P_1 and P_2 be 16, α, β, then the value of α + β is equal to ____. **[JEE (Main) 2022]**

MTD044

20. If the two lines $l_1:\frac{x-2}{3}=\frac{y+1}{-2}$, z = 2 and $l_2:\frac{x-1}{1}=\frac{2y+3}{\alpha}=\frac{z+5}{2}$ perpendicular, then an angle between the lines l_2 and $l_3:\frac{1-x}{3}=\frac{2y-1}{-4}=\frac{z}{4}$ is : **[JEE (Main) 2022]**

(1) $\cos^{-1}\left(\frac{29}{4}\right)$ (2) $\sec^{-1}\left(\frac{29}{4}\right)$ (3) $\cos^{-1}\left(\frac{2}{29}\right)$ (4) $\cos^{-1}\left(\frac{2}{\sqrt{29}}\right)$

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MTD045

21. Let N be the foot of perpendicular from the point $P(1,-2,3)$ on the line passing through the points (4, 5, 8) and (1, −7, 5). Then the distance of N from the plane $2x-2y+z+5=0$ is **[JEE (Main) 2023]**

(1) 6 (2) 9 (3) 7 (4) 8

MTD124

22. If the foot of the perpendicular drawn from (1, 9, 7) to the line passing through the point (3, 2, 1) and parallel to the planes $x+2y+z=0$ and $3y-z=3$ is (α, β, γ), then α + β + γ is equal to **[JEE (Main) 2023]**

(1) −1 (2) 3 (3) 1 (4) 5

MTD125

23. One vertex of a rectangular parallelopiped is at the origin O and the lengths of its edges along x, y and z axes are 3, 4 and 5 units respectively. Let P be the vertex (3, 4, 5). Then the shortest distance between the diagonal OP and an edge parallel to z axis, not passing through O or P is: **[JEE (Main) 2023]**

(1) $\frac{12}{\sqrt{5}}$ (2) $\frac{12}{5\sqrt{5}}$ (3) $12\sqrt{5}$ (4) $\frac{12}{5}$

MTD126

For More Join: @IITJEE_Advanced

24. The line l_1 passes through the point (2,6,2) and is perpendicular to the plane $2x + y - 2z = 10$. Then the shortest distance between the line l_1 and the line $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ is :

[JEE (Main) 2023]

(1) 7 (2) $\frac{19}{3}$ (3) $\frac{19}{3}$ (4) 9

MTD127

25. Let the plane $x + 3y - 2z + 6 = 0$ meet the co-ordinate axes at the points A, B, C. If the orthocentre of the triangle ABC is $\left(\alpha, \beta, \frac{6}{7}\right)$, then $98\,(\alpha + \beta)^2$ is equal to_____. **[JEE (Main) 2023]**

MTD128

26. If $\lambda_1 < \lambda_2$ are two values of λ such that the angle between the planes $P_1 : \vec{r}\left(3\hat{i} - 5j + k\right) = 7$ and $P_2 : \vec{r} \cdot \left(\lambda\hat{i} + j - 3k\right) = 9$ is $\sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$, then the square of the length of perpendicular from the point $(38\lambda_1, 10\lambda_2, 2)$ to the plane P_1 is ____. **[JEE (Main) 2023]**

MTD129

27. Let θ be the angle between the planes $P_1 = \vec{r}.\left(\hat{i} + \hat{j} + 2\hat{k}\right) = 9$ and $P_2 = \vec{r}.\left(2\hat{i} - \hat{j} + \hat{k}\right) = 15$. Let L be the line that meets P_2 at the point (4, –2, 5) and makes an angle θ with the normal of P_2. If α is the angle between L and P_2 then $(\tan^2\theta)(\cot^2\alpha)$ is equal to _____. **[JEE (Main) 2023]**

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MTD130

28. Let a line L pass through the point $P(2, 3, 1)$ and be parallel to the line $x + 3y - 2z - 2 = 0 = x - y + 2z$. If the distance of L from the point (5, 3, 8) is α, then $3\alpha^2$ is equal to ____. **[JEE (Main) 2023]**

MTD131

29. Let the plane $P : 8x + \alpha_1 y + \alpha_2 z + 12 = 0$ be parallel to the line $L : \frac{x+2}{2} = \frac{y-3}{3} = \frac{z+4}{5}$. If the intercept of P on the y-axis is 1, then the distance between P and L is : **[JEE (Main) 2023]**

(1) $\sqrt{14}$ (2) $\frac{6}{\sqrt{14}}$ (3) $\sqrt{\frac{2}{7}}$ (4) $\sqrt{\frac{7}{2}}$

MTD132

30. Let the plane P pass through the intersection of the planes $2x + 3y - z = 2$ and $x + 2y + 3z = 6$, and be perpendicular to the plane $2x + y - z + 1 = 0$. If d is the distance of P from the point $(-7,1,1)$, then d^2 is equal to : **[JEE (Main) 2023]**

(1) $\frac{250}{83}$ (2) $\frac{15}{53}$ (3) $\frac{25}{83}$ (4) $\frac{250}{82}$

MTD133

For More Join: @IITJEE_Advanced

31. Let P the point of intersection of the lines $\frac{x-2}{1}=\frac{y-4}{5}=\frac{z-2}{1}$ and $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-3}{2}$. Then, the shortest distance of P from the line $4x = 2y = z$ is **[JEE (Main) 2024]**

(1) $\frac{5\sqrt{14}}{7}$ (2) $\frac{\sqrt{14}}{7}$ (3) $\frac{3\sqrt{14}}{7}$ (4) $\frac{6\sqrt{14}}{7}$

MTD205

32. Consider a line L passing through the points $P(1,2,1)$ and $Q(2,1,-1)$. If the mirror image of the point $A(2,2,2)$ in the line L is (α,β,γ), then $\alpha + \beta + 6\gamma$ is equal to ..... **[JEE (Main) 2024]**

MTD206

33. If the shortest distance between the lines $\frac{x+2}{2}=\frac{y+3}{3}=\frac{z-5}{4}$ and $\frac{x-3}{1}=\frac{y-2}{-3}=\frac{z+4}{2}$ is $\frac{38}{3\sqrt{5}}k$ and $\int_0^k [x^2]dx=\alpha-\sqrt{\alpha}$, where $[x]$ denotes the greatest integer function, then $6\alpha^3$ is equal to ____ **[JEE (Main) 2024]**

MTD207

34. Let the point $(-1, \alpha, \beta)$ lie on the line of the shortest distance between the lines $\frac{x+2}{-3}=\frac{y-2}{4}=\frac{z-5}{2}$ and $\frac{x+2}{-1}=\frac{y+6}{2}=\frac{z-1}{0}$. Then $(\alpha - \beta)^2$ is equal to ____. **[JEE (Main) 2024]**

MTD208

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35. Let d be the distance of the point of intersection of the lines $\frac{x+6}{3}=\frac{y}{2}=\frac{z+1}{1}$ and $\frac{x-7}{4}=\frac{y-9}{3}=\frac{z-4}{2}$ from the point (7, 8, 9). Then $d^2 + 6$ is equal to : **[JEE (Main) 2024]**

(1) 72 (2) 69 (3) 75 (4) 78

MTD209

36. If the line $\frac{2-x}{3}=\frac{3y-2}{4\lambda+1}=4-z$ makes a right angle with the line $\frac{x+3}{3\mu}=\frac{1-2y}{6}=\frac{5-z}{7}$, then $4\lambda + 9\mu$ is equal to : **[JEE (Main) 2024]**

(1) 13 (2) 4 (3) 5 (4) 6

MTD210

37. Let $P(\alpha, \beta, \gamma)$ be the image of the point $Q(3, -3, 1)$ in the line $\frac{x-0}{1}=\frac{y-3}{1}=\frac{z-1}{-1}$ and R be the point $(2, 5, -1)$. If the area of the triangle PQR is λ and $\lambda^2 = 14K$, then K is equal to: **[JEE (Main) 2024]**

(1) 36 (2) 72 (3) 18 (4) 81

MTD211

38. Let P be the point $(10, -2, -1)$ and Q be the foot of the perpendicular drawn from the point $R(1, 7, 6)$ on the line passing through the points $(2, -5, 11)$ and $(-6, 7, -5)$. Then the length of the line segment PQ is equal to _____. **[JEE (Main) 2024]**

MTD212

For More Join: @IITJEE_Advanced

39. Let $P(x, y, z)$ be a point in the first octant, whose projection in the xy-plane is the point Q. Let $OP = \gamma$; the angle between OQ and the positive x-axis be θ; and the angle between OP and the positive z-axis be ϕ, where O is the origin. Then the distance of P from the x-axis is :

[JEE (Main) 2024]

(1) $\gamma\sqrt{1-\sin^2\phi\cos^2\theta}$ (2) $\gamma\sqrt{1+\cos^2\theta\sin^2\phi}$ (3) $\gamma\sqrt{1-\sin^2\theta\cos^2\phi}$ (4) $\gamma\sqrt{1+\cos^2\phi\sin^2\theta}$

MTD213

40. Let the line L intersect the lines

$x - 2 = -y = z - 1$, $2(x + 1) = 2(y - 1) = z + 1$

and be parallel to the line $\frac{x-2}{3}=\frac{y-1}{1}=\frac{z-2}{2}$. Then which of the following points lies on L ?

[JEE (Main) 2024]

(1) $\left(-\frac{1}{3},1,1\right)$ (2) $\left(-\frac{1}{3},1,-1\right)$ (3) $\left(-\frac{1}{3},-1,-1\right)$ (4) $\left(-\frac{1}{3},-1,1\right)$

MTD214

41. The shortest distance between the line

$\frac{x-3}{4}=\frac{y+7}{-11}=\frac{z-1}{5}$ and $\frac{x-5}{3}=\frac{y-9}{-6}=\frac{z+2}{1}$ is :

[JEE (Main) 2024]

(1) $\frac{187}{\sqrt{563}}$ (2) $\frac{178}{\sqrt{563}}$ (3) $\frac{185}{\sqrt{563}}$ (4) $\frac{179}{\sqrt{563}}$

MTD215

42. Let the image of the point (1, 0, 7) in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ be the point (α, β, γ). Then which one of the following points lies on the line passing through (α, β, γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with y-axis and z-axis respectively and an acute angle with x-axis ?

[JEE (Main) 2024]

(1) $\left(1,-2,1+\sqrt{2}\right)$ (2) $\left(1,2,1-\sqrt{2}\right)$ (3) $\left(3,4,3-2\sqrt{2}\right)$ (4) $\left(3,-4,3+2\sqrt{2}\right)$

MTD216

43. The lines $\frac{x-2}{2}=\frac{y}{-2}=\frac{z-7}{16}$ and $\frac{x+3}{4}=\frac{y+2}{3}=\frac{z+2}{1}$ intersect at the point P. If the distance of P from the line $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-1}{1}$ is l, then $14l^2$ is equal to.....

[JEE (Main) 2024]

MTD217

44. A line with direction ratios 2, 1, 2 meets the lines $x = y + 2 = z$ and $x + 2 = 2y = 2z$ respectively at the point P and Q. if the length of the perpendicular from the point (1, 2, 12) to the line PQ is l, then l^2 is

[JEE (Main) 2024]

MTD218

45. Let a line passing through the point (–1, 2, 3) intersect the lines $L_1:\frac{x-1}{3}=\frac{y-2}{2}=\frac{z+1}{-2}$ at $M(\alpha,\beta,\gamma)$ and $L_2:\frac{x+2}{-3}=\frac{y-2}{-2}=\frac{z-1}{4}$ at $N(a, b, c)$. Then the value of $\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}$ equals ___.

[JEE (Main) 2024]

MTD219

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For More Join: @IITJEE_Advanced

46. The distance of the point $Q(0, 2, -2)$ form the line passing through the point $P(5, -4, 3)$ and perpendicular to the lines $\vec{r}=\left(-3\hat{i}+2\hat{k}\right)+\lambda\left(2\hat{i}+3\hat{j}+5\hat{k}\right)$, $\lambda\in\mathbb{R}$ and $\vec{r}=\left(\hat{i}-2\hat{j}+\hat{k}\right)+\mu\left(-\hat{i}+3\hat{j}+2\hat{k}\right)$, $\mu\in\mathbb{R}$ is **[JEE (Main) 2024]**

(1) $\sqrt{86}$ (2) $\sqrt{20}$ (3) $\sqrt{54}$ (4) $\sqrt{74}$

MTD220

47. Let P and Q be the points on the line $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ which are at a distance of 6 units from the point $R(1,2,3)$. If the centroid of the triangle PQR is (α, β,γ), then $\alpha^2+\beta^2+\gamma^2$ is: **[JEE (Main) 2024]**

(1) 26 (2) 36 (3) 18 (4) 24

MTD221

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EXERCISE - JEE (Advanced) PYQ

1. Perpendiculars are drawn from points on the line $\frac{x+2}{2}=\frac{y+1}{-1}=\frac{z}{3}$ to the plane $x+y+z=3$. The feet of perpendiculars lie on the line **[JEE (Advanced) 2013]**

(A) $\frac{x}{5}=\frac{y-1}{8}=\frac{z-2}{-13}$ (B) $\frac{x}{2}=\frac{y-1}{3}=\frac{z-2}{-5}$ (C) $\frac{x}{4}=\frac{y-1}{3}=\frac{z-2}{-7}$ (D) $\frac{x}{2}=\frac{y-1}{-7}=\frac{z-2}{5}$

MTD064

2. A line ℓ passing through the origin is perpendicular to the lines

$\ell_1:(3+t)\hat{i}+(-1+2t)\hat{j}+(4+2t)\hat{k},-\infty<t<\infty$

$\ell_2:(3+2s)\hat{i}+(3+2s)\hat{j}+(2+s)\hat{k},-\infty<s<\infty$

Then, the coordinate(s) of the point(s) on ℓ_2 at a distance of $\sqrt{17}$ from the point of intersection of ℓ and ℓ_1 is(are) - **[JEE (Advanced) 2013]**

(A) $\left(\frac{7}{3},\frac{7}{3},\frac{5}{3}\right)$ (B) (–1,–1,0) (C) (1,1,1) (D) $\left(\frac{7}{9},\frac{7}{9},\frac{8}{9}\right)$

MTD065

3. Two lines $L_1:x=5,\frac{y}{3-\alpha}=\frac{z}{-2}$ and $L_2:x=\alpha,\frac{y}{-1}=\frac{z}{2-\alpha}$ are coplanar. Then α can take value(s)

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[JEE (Advanced) 2013]

(A) 1 (B) 2 (C) 3 (D) 4

MTD066

4. Consider the lines $L_1:\frac{x-1}{2}=\frac{y}{-1}=\frac{z+3}{1},L_2:\frac{x-4}{1}=\frac{y+3}{1}=\frac{z+3}{2}$ and the planes $P_1:7x+y+2z=3$, $P_2:3x+5y-6z=4$. Let $ax+by+cz=d$ be the equation of the plane passing through the point of intersection of lines L_1 and L_2 and perpendicular to planes P_1 and P_2.

Match **List-I** with **List-II** and select the correct answer using the code given below the lists.

List-I	**List-II**
(P) a =	1. 13
(Q) b =	2. –3
(R) c =	3. 1
(S) d =	4. –2

[JEE (Advanced) 2013]

Codes :

	P	**Q**	**R**	**S**
(A)	3	2	4	1
(B)	1	3	4	2
(C)	3	2	1	4
(D)	2	4	1	3

MTD067

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5. From a point $P(\lambda,\lambda,\lambda)$, perpendiculars PQ and PR are drawn respectively on the lines $y = x, z = 1$ and $y = -x, z = -1$. If P is such that $\angle QPR$ is a right angle, then the possible value(s) of λ is(are)

[JEE (Advanced) 2014]

(A) $\sqrt{2}$ (B) 1 (C) –1 (D) $-\sqrt{2}$

MTD068

6. In $\mathbb{R}^3$, consider the planes $P_1: y = 0$ and $P_2: x + z = 1$. Let P_3 be a plane, different from P_1 and P_2, which passes through the intersection of P_1 and P_2. If the distance of the point (0,1,0) from P_3 is 1 and the distance of a point (α,β,γ) from P_3 is 2, then which of the following relations is (are) true ?

[JEE (Advanced) 2015]

(A) $2\alpha + \beta + 2\gamma + 2 = 0$ (B) $2\alpha - \beta + 2\gamma + 4 = 0$

(C) $2\alpha + \beta - 2\gamma - 10 = 0$ (D) $2\alpha - \beta + 2\gamma - 8 = 0$

MTD069

7. In $\mathbb{R}^3$, let L be a straight line passing through the origin. Suppose that all the points on L are at a constant distance from the two planes $P_1: x + 2y - z + 1 = 0$ and $P_2: 2x - y + z - 1 = 0$. Let M be the locus of the feet of the perpendiculars drawn from the points on L to the plane P_1. Which of the following points lie(s) on M ?

[JEE (Advanced) 2015]

(A) $\left(0,-\frac{5}{6},-\frac{2}{3}\right)$ (B) $\left(-\frac{1}{6},-\frac{1}{3},\frac{1}{6}\right)$ (C) $\left(-\frac{5}{6},0,\frac{1}{6}\right)$ (D) $\left(-\frac{1}{3},0,\frac{2}{3}\right)$

MTD070

8. Consider a pyramid $OPQRS$ located in the first octant $(x \geq 0, y \geq 0, z \geq 0)$ with O as origin, and OP and OR along the x –axis and the y –axis, respectively. The base $OPQR$ of the pyramid is a square with $OP = 3$. The point S is directly above the mid-point T of diagonal OQ such that $TS = 3$. Then-

[JEE (Advanced) 2016]

(A) the acute angle between OQ and OS is $\frac{\pi}{3}$.

(B) the equation of the plane containing the triangle OQS is $x - y = 0$

(C) the length of the perpendicular from P to the plane containing the triangle OQS is $\frac{3}{\sqrt{2}}$

(D) the perpendicular distance from O to the straight line containing RS is $\sqrt{\frac{15}{2}}$

MTD071

9. Let P be the image of the point $(3, 1, 7)$ with respect to the plane $x - y + z = 3$. Then the equation of the plane passing through P and containing the straight line $\frac{x}{1}=\frac{y}{2}=\frac{z}{1}$ is

[JEE (Advanced) 2016]

(A) $x + y - 3z = 0$ (B) $3x + z = 0$ (C) $x - 4y + 7z = 0$ (D) $2x - y = 0$

MTD072

10. The equation of the plane passing through the point (1,1,1) and perpendicular to the planes $2x + y - 2z = 5$ and $3x - 6y - 2z = 7$, is-

[JEE (Advanced) 2017]

(A) $14x + 2y + 15z = 31$ (B) $14x + 2y - 15z = 1$

(C) $-14x + 2y + 15z = 3$ (D) $14x - 2y + 15z = 27$

MTD073

For More Join: @IITJEE_Advanced

11. Let $P_1: 2x + y - z = 3$ and $P_2: x + 2y + z = 2$ be two planes. Then, which of the following statement(s) is (are) TRUE ? **[JEE (Advanced) 2018]**

(A) The line of intersection of P_1 and P_2 has direction ratios $1, 2, -1$

(B) The line $\frac{3x-4}{9} = \frac{1-3y}{9} = \frac{z}{3}$ is perpendicular to the line of intersection of P_1 and P_2

(C) The acute angle between P_1 and P_2 is $60°$

(D) If P_3 is the plane passing through the point (4, 2, –2) and perpendicular to the line of intersection of P_1 and P_2, then the distance of the point (2, 1, 1) from the plane P_3 is $\frac{2}{\sqrt{3}}$

MTD074

12. Let P be a point in the first octant, whose image Q in the plane $x + y = 3$ (that is, the line segment PQ is perpendicular to the plane $x + y = 3$ and the mid-point of PQ lies in the plane $x + y = 3$) lies on the z –axis. Let the distance of P from the x –axis be 5. If R is the image of P in the xy –plane, then the length of PR is ____ . **[JEE (Advanced) 2018]**

MTD075

13. Consider the cube in the first octant with sides OP, OQ and OR of length 1, along the x –axis, y –axis and z –axis, respectively, where $O(0, 0, 0)$ is the origin. Let $S\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$ be the centre of the cube and T be the vertex of the cube opposite to the origin O such that S lies on the diagonal OT. If $\vec{p} = \overrightarrow{SP}$, $\vec{q} = \overrightarrow{SQ}$, $\vec{r} = \overrightarrow{SR}$ and $\vec{t} = \overrightarrow{ST}$, then the value of $\left|(\vec{p} \times \vec{q}) \times (\vec{r} \times \vec{t})\right|$ is ____ . **[JEE (Advanced) 2018]**

MTD076

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14. Let L_1 and L_2 denotes the lines $\vec{r} = \hat{i} + \lambda(-\hat{i} + 2\hat{j} + 2\hat{k}), \lambda \in \mathbb{R}$ and $\vec{r} = \mu(2\hat{i} - \hat{j} + 2\hat{k}), \mu \in \mathbb{R}$ respectively. If L_3 is a line which is perpendicular to both L_1 and L_2 and cuts both, then which of the following options describe(s) L_3 ? **[JEE (Advanced) 2019]**

(A) $\vec{r} = \frac{1}{3}(2\hat{i} + \hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in \mathbb{R}$

(B) $\vec{r} = \frac{2}{9}(2\hat{i} - \hat{j} + 2\hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in \mathbb{R}$

(C) $\vec{r} = t(2\hat{i} + 2\hat{j} - \hat{k}), t \in \mathbb{R}$

(D) $\vec{r} = \frac{2}{9}(4\hat{i} + \hat{j} + \hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in \mathbb{R}$

MTD077

15. Three lines are given by

$\vec{r} = \lambda\,\hat{i}, \lambda \in \mathbb{R}$

$\vec{r} = \mu(\hat{i} + \hat{j}), \mu \in \mathbb{R}$ and

$\vec{r} = \nu(\hat{i} + \hat{j} + \hat{k}), \nu \in \mathbb{R}$.

Let the lines cut the plane $x + y + z = 1$ at the points A, B and C respectively. If the area of the triangle ABC is Δ then the value of $(6\Delta)^2$ equals ___ **[JEE (Advanced) 2019]**

MTD078

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16. Let L_1 and L_2 be the following straight lines.

$L_1 : \frac{x-1}{1} = \frac{y}{-1} = \frac{z-1}{3}$ and $L_2 : \frac{x-1}{-3} = \frac{y}{-1} = \frac{z-1}{1}$

Suppose the straight line

$L : \frac{x-\alpha}{l} = \frac{y-1}{m} = \frac{z-\gamma}{-2}$

lies in the plane containing L_1 and L_2, and passes through the point of intersection of L_1 and L_2. If the line L bisects the acute angle between the lines L_1 and L_2, then which of the following statements is/are TRUE? **[JEE (Advanced) 2020]**

(A) $\alpha - \gamma = 3$ (B) $\ell + m = 2$ (C) $\alpha - \gamma = 1$ (D) $\ell + m = 0$

MTD079

17. Let $\alpha, \beta, \gamma, \delta$ be real numbers such that $\alpha^2 + \beta^2 + \gamma^2 \neq 0$ and $\alpha + \gamma = 1$. Suppose the point $(3,2,-1)$ is the mirror image of the point $(1,0,-1)$ with respect to the plane $\alpha x + \beta y + \gamma z = \delta$. Then which of the following statements is/are TRUE ? **[JEE (Advanced) 2020]**

(A) $\alpha + \beta = 2$ (B) $\delta - \gamma = 3$ (C) $\delta + \beta = 4$ (D) $\alpha + \beta + \gamma = \delta$

MTD080

Paragraph for Question No. 18 to 19

Let α, β and γ be real numbers such that the system of linear equations

$x + 2y + 3z = \alpha$

$4x + 5y + 6z = \beta$

$7x + 8y + 9z = \gamma - 1$

is consistent. Let $|M|$ represent the determinant of the matrix $M = \begin{bmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

Let P be the plane containing all those (α, β, γ) for which the above system of linear equations is consistent, and D be the square of the distance of the point $(0, 1, 0)$ from the plane P.

18. The value of $|M|$ is _____. **[JEE (Advanced) 2021]**

MTD081

19. The value of D is _____.

[JEE (Advanced) 2021]

MTD082

20. Let P_1 and P_2 be two planes given by

P_1: $10x + 15y + 12z - 60 = 0$,

$P_2 : -2x + 5y + 4z - 20 = 0$.

Which of the following straight lines can be an edge of some tetrahedron whose two faces lie on P_1 and P_2 ? **[JEE (Advanced) 2022]**

(A) $\frac{x-1}{0} = \frac{y-1}{0} = \frac{z-1}{5}$ (B) $\frac{x-6}{-5} = \frac{y}{2} = \frac{z}{3}$

(C) $\frac{x}{-2} = \frac{y-4}{5} = \frac{z}{4}$ (D) $\frac{x}{1} = \frac{y-4}{-2} = \frac{z}{3}$

MTD083

For More Join: @IITJEE_Advanced

21. Let Q be the cube with the set of vertices $\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1, x_2, x_3 \{0,1\}\}$. Let F be the set of all twelve lines containing the diagonals of the six faces of the cube Q. Let S be the set of all four lines containing the main diagonals of the cube Q; for instance, the line passing through the vertices (0, 0, 0) and (1, 1, 1) is in S. For lines ℓ_1 and ℓ_2, let $d(\ell_1, \ell_2)$ denote the shortest distance between them. Then the maximum value of d(ℓ_1, ℓ_2), as ℓ_1 varies over F and ℓ_2 varies over S, is

[JEE (Advanced) 2023]

(A) $\frac{1}{\sqrt{6}}$ (B) $\frac{1}{\sqrt{8}}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{\sqrt{12}}$

MTD134

22. Let P be the plane $\sqrt{3}x+2y+3z=16$ and let

$S=\left\{\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k} : \alpha^2+\beta^2+\gamma^2=1 \text{ and the distance of } (\alpha,\beta,\gamma) \text{ from the plane } P \text{ is } \frac{7}{2}\right\}$.

Let $\vec{u}, \vec{v}$ and $\vec{w}$ be three distinct vectors in S such that $|\vec{u}-\vec{v}|=|\vec{v}-\vec{w}|=|\vec{w}-\vec{u}|$. Let V be the volume of the parallelepiped determined by vectors $\vec{u}, \vec{v}$ and $\vec{w}$. Then the value of $\frac{80}{\sqrt{3}}V$ is

[JEE (Advanced) 2023]

MTD135

23. Let ℓ_1 and ℓ_2 be the lines $\vec{r}_1=\lambda(\hat{i}+\hat{j}+\hat{k})$ and $\vec{r}_2=(\hat{j}-\hat{k})+\mu(\hat{i}+\hat{k})$, respectively. Let X be the set of all the planes H that contain the line ℓ_1. For a plane H, let $d(H)$ denote the smallest possible distance between the points of ℓ_2 and H. Let H_0 be plane in X for which $d(H_0)$ is the maximum value of $d(H)$ as H varies over all planes in X.

Match each entry in **List-I** to the correct entries in **List-II.**

List-I		**List-II**	
(P)	The value of $d(H_0)$ is	(1)	$\sqrt{3}$
(Q)	The distance of the point (0,1,2) from H_0 is	(2)	$\frac{1}{\sqrt{3}}$
(R)	The distance of origin from H_0 is	(3)	0
(S)	The distance of origin from the point of intersection of planes $y = z, x = 1$ and H_0 is	(4)	$\sqrt{2}$
		(5)	$\frac{1}{\sqrt{2}}$

The correct option is :

[JEE (Advanced) 2023]

(A) (P) → (2) (Q) → (4) (R) → (5) (S) → (1)
(B) (P) → (5) (Q) → (4) (R) → (3) (S) → (1)
(C) (P) → (2) (Q) → (1) (R) → (3) (S) → (2)
(D) (P) → (5) (Q) → (1) (R) → (4) (S) → (2)

MTD136

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24. Let $\mathbb{R}^3$ denote the three-dimensional space. Take two points P = (1, 2, 3) and Q = (4, 2, 7). Let dist (X, Y) denote the distance between two points X and Y in $\mathbb{R}^3$. Let

$S = \left\{X \in \mathbb{R}^3 : (\text{dist}(X, P))^2 - (\text{dist}(X, Q))^2 = 50\right\}$ and

$T = \left\{Y \in \mathbb{R}^3 : (\text{dist}(Y, Q))^2 - (\text{dist}(Y, P))^2 = 50\right\}$.

Then which of the following statements is (are) TRUE ? **[JEE (Advanced) 2024]**

(A) There is a triangle whose area is 1 and all of whose vertices are from S.

(B) There are two distinct points L and M in T such that each point on the line segment LM is also in T.

(C) There are infinitely many rectangles of perimeter 48, two of whose vertices are from S and the other two vertices are from T.

(D) There is a square of perimeter 48, two of whose vertices are from S and the other two vertices are from T.

MTD222

25. Let $\gamma \in \mathbb{R}$ be such that the lines $L_1 : \frac{x+11}{1} = \frac{y+21}{2} = \frac{z+29}{3}$ and $L_2 : \frac{x+16}{3} = \frac{y+11}{2} = \frac{z+4}{\gamma}$ intersect.

Let R_1 be the point of intersection of L_1 and L_2. Let O = (0, 0, 0), and $\hat{n}$ denote a unit normal vector to the plane containing both the lines L_1 and L_2.

Match each entry in **List-I** to the correct entry in **List-II**.

	List-I		**List-II**
(P)	γ equals	(1)	$-\hat{i} - \hat{j} + \hat{k}$
(Q)	A possible choice for $\hat{n}$ is	(2)	$\sqrt{\frac{3}{2}}$
(R)	$\overrightarrow{OR_1}$ equals	(3)	1
(S)	A possible value of $\overrightarrow{OR_1}.\hat{n}$ is	(4)	$\frac{1}{\sqrt{6}}\hat{i} - \frac{2}{\sqrt{6}}\hat{j} + \frac{1}{\sqrt{6}}\hat{k}$
		(5)	$\sqrt{\frac{2}{3}}$

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The correct option is **[JEE (Advanced) 2024]**

(A) (P) → (3) (Q) → (4) (R) → (1) (S) → (2)

(B) (P) → (5) (Q) → (4) (R) → (1) (S) → (2)

(C) (P) → (3) (Q) → (4) (R) → (1) (S) → (5)

(D) (P) → (3) (Q) → (1) (R) → (4) (S) → (5)

MTD223

26. A straight line drawn from the point $P(1, 3, 2)$, parallel to the line $\frac{x-2}{1} = \frac{y-4}{2} = \frac{z-6}{1}$, intersects the plane $L_1 : x - y + 3z = 6$ at the point Q. Another straight line which passes through Q and is perpendicular to the plane L_1 intersects the plane $L_2 : 2x - y + z = -4$ at the point R.

Then which of the following statements is (are) TRUE? **[JEE (Advanced) 2024]**

(A) The length of the line segment PQ is $\sqrt{6}$

(B) The coordinates of R are (1, 6, 3)

(C) The centroid of the triangle PQR is $\left(\frac{4}{3}, \frac{14}{3}, \frac{5}{3}\right)$

(D) The perimeter of the triangle PQR is $\sqrt{2} + \sqrt{6} + \sqrt{11}$

MTD224

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	D	A	B	D	B	A	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	C	C	D	A	C	B	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	C	B	C	A	B	A	A	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,D	A,D	A,B,C	A,C	A,C	A,B,D	A,D	C,D	B,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,C	B,C,D	B,D	C,D	B,C,D	A,B,C,D	B,C	B,C	B	A,B
Que.	21	22	23	24	25	26	27	28	29	
Ans.	C	D	C	A	A,B,C	A,B,C	1.41 or 1.42	0.66 or 0.67	D	
Que.	30									
Ans.	A → R; B → Q; C → P; D → S									

EXERCISE (S)

1. 3 **2.** 17 **3.** 6 **4.** 9 **5.** 6 **6.** 3
7. 4.50 **8.** 3.50 **9.** 4.00 **10.** 2.44 or 2.45

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	4.00	2	1	3	28	2	4	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	84	1	1	1	2	4	2	4	28	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	4	4	4	288	315	9	158	1	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	3	6	48	25	3	4	4	13	1	2
Que.	41	42	43	44	45	46	47			
Ans.	1	3	108	65	196	4	3			

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B,D	A,D	A	C	B,D	A,B	B,C,D	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C,D	8	0.5	A,B,D	0.75	A,B	A,B,C	1.00	1.50	A,B,D
Que.	21	22	23	24	25	26				
Ans.	A	45	B	A,B,C,D	C	A,C				

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